

Title: Modern Science Outreach: A Guide for Scientists

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Summary: Science communication offers immense value for fostering public trust and enhancing the quality of our modern information ecosystem. However, it remains under-prioritized and is rarely taught in the scientific enterprise. We thus face an enormous opportunity to address this gap by upskilling scientists in public engagement efforts. To that end, this paper outlines six actionable practices to help scientists effectively reach and engage broad audiences, with a focus on leveraging modern media and emerging dissemination channels. Notably, these practices bridge theory and practice: they are both theoretically backed and have been successfully implemented already by the authors. We encourage scientists at all levels, and across disciplines, to implement these strategies to engage broad audiences and contribute positively to public discourse.

Keywords:

Science communication, open science, accessibility, messaging, public understanding, social media, podcasts

Main Text

Introduction:

Science communication is a collaborative exchange¹ that translates evidence-based knowledge into accessible and trustworthy forms through ongoing conversation. It is a rapidly growing field with major implications for society's relationship with science, as it allows scientific information to be culturally integrated, debated, and reshaped by diverse voices of society.² Today's media landscape is characterized by a new, democratized communications model where individuals with large social media followings are challenging traditional institutions and broadcasters as the primary holders of influence.³ This new model permits more participatory science communication strategies - as audiences may engage directly with communicators by commenting or messaging – thereby supporting modern theoretical models of science communication as a dialogue rather than just knowledge transfer.⁴ Many active and formerly active scientists have built careers around science communication, discussing research findings and other materials with the public through books,⁵ TV shows,⁶ blogs,⁷ podcasts,^{8,9} social media posts,¹⁰ and other forms of traditional and non-traditional media.

While peer-reviewed publications remain the backbone of the scientific enterprise – exemplified by *Nature's* roughly nine million monthly readers¹¹ - their reach and impact are limited by technical jargon and paywalls. In contrast, social media use has skyrocketed over the past decade, rising to an estimated 4.9 billion users as of 2023.¹² Facebook alone is used by 2 billion people *every day*.¹³ These platforms are optimized for attention, and recent data suggest that a large proportion (72%¹⁴) of users rely on social media as a source of news. Scientists are largely underrepresented on these platforms, leaving a vacuum often filled by non-experts. This imbalance may be facilitating the spread of misinformation and contributing to the documented decline in public trust in science.¹⁵ One promising avenue for rebuilding trust is framing science as personally relevant and close (rather than abstract and distant) which has been shown to reduce skepticism and increase positive attitudes toward science.¹⁶ The scientific community must take responsibility for sharing research in accessible formats and develop routes for connecting with the public.¹⁷ To achieve this, we must engage and upskill scientists on communicating through available media channels, of which there are several established and ready for deployment.

Increasingly the visibility and presence of scientists in modern public discourse offers a vital mechanism for enabling conversations between science and society. By engaging in direct dialogue – answering questions and acknowledging diverse perspectives – science communicators can humanize the scientific process and get people excited about science.¹⁸ Given the prevalence of medical misinformation¹⁹ and the fact that 83% of the public expresses a desire to hear from scientists more about

their work,²⁰ there is an urgent need for science communicators. However, not all scientists have the training or resources to function as science communicators. Thus, this piece offers a guide to six science communication approaches that fall within different ranges of efforts and are theoretically backed, having been put into practice already.

The Audience

Before implementing any of the practices proposed below, it is critical to understand *who* you are trying to reach. The effectiveness of any communication strategy depends on a clear grasp of your priority audience. The “public” is rarely homogenous; audiences are composed of individuals with distinct identities, beliefs, experiences, and motivations.²¹ What resonates with one group may fall flat with another, so thoughtful audience analysis is critical. Audience analysis²² involves a structured process to help you understand your audience and design messages that will resonate.²³ There are three main steps:

1. **Identify your priority audience and secondary audiences.** Start by brainstorming all potential audiences connected to your communication goal. Then narrow your focus to the priority audience—the group whose behavior you must influence to achieve your goal. Secondary audiences are those who influence the priority audience’s behaviors, positively or negatively. For example, if your goal is to inspire students to pursue STEM careers, high school students may be your priority audience, while teachers and guidance counselors serve as influential secondary audiences.
2. **Understand your audience’s characteristics.** Dive deeper into the demographic, geographic, psychographic, and communication preferences of your priority audience. Demographics provide basic descriptors like age or education, while psychographics capture values, interests, and motivations. Communication preferences help you identify the channels and contexts where your audience is most likely to engage with your message. Issue-specific beliefs and behaviors reveal what your audience already knows, believes, or does in relation to your communication goal. The richer your understanding, the better you can design messages that land.
3. **Identify barriers to engagement.** Consider the factors that may prevent your audience from adopting your message or behavior. These can include negative experiences, misconceptions, fear, entrenched habits, or social norms. Understanding these barriers allows you to anticipate resistance and tailor your communication to address or circumvent these challenges effectively.

A detailed understanding of audience allows communicators to “meet people where they are,” designing messages that align with the audience’s perspectives, needs, and emotional context. This

process is not about changing who your audience is but about tailoring your approach to maximize engagement and impact. Even small efforts toward audience understanding can significantly enhance the success of your communication practices.

Six Modern Outreach Practices

Here we outline six actionable science communication strategies that bridge theory and practice. They are intended to broaden the existing definition of outreach to include modern media channels and emerging digital routes for engaging audiences with science. Recognizing that scientific careers are demanding, we have categorized these practices by their required investment and potential impact to accommodate researchers with limited bandwidth. Beneath each item, there will be two scores: Potential Reach (PR; **Fig. 1**) and Effort/Time Investment (ETI; **Fig. 2**), each ranging from 1-5. These scores are meant to illustrate the reach and value of each activity along with the effort and time commitment it requires.

#1 - CRAFT YOUR RESEARCH SUMMARY

Effort/Time Investment: *1 - One time commitment of <1 day of work.*

Potential Reach: *Groups 1, 2 and 3 – Scientific community, personal network, and local community.*

Imagine you're at an airport bar waiting for a flight, and the stranger beside you strikes up a conversation. As the discussion naturally carries into occupation, they ask about your research. While they admit they're no expert, they're genuinely curious about your research topic. They would love to hear about your research in simple terms. Are you prepared to do this?

An important first consideration is audience. As indicated above, effective science communication is a dialogue, and it is critical to understand who you are interacting with. What is their occupation? What level of science education have they received? What perceived barriers might they have to science, such as bad previous experiences, sociocultural influences, or misconceptions? Your approach to the conversation – such as the language you use, what aspects to highlight, and what analogies may be required - will likely differ based on each of these factors. Being able to recognize the disposition of your audience is just as important as crafting an effective research summary.

Every researcher should have a concise, accessible summary of their research prepared, often referred to as an “elevator pitch.” We have a responsibility to support public literacy of science by speaking accessibly, accurately, and transparently. Whether it's for an uncle, a friend, or the curious airport bar patron, it's important to be prepared to discuss science in a lay-friendly way. Communicating

within our communities is the most fundamental form of science communication, as it breaks down barriers and fosters trust in science in casual settings.

It is thus important to devise an accurate, clear, and simple explanation of your work. This can be done in a single sitting. Dedicate some time, approximately 30-60 minutes, to consider your research from the broadest scope imaginable. What do you do, and why? What are your research goals? What is the ideal outcome? How have you advanced toward that outcome? How will your work impact humanity either at the individual or macro level? What biological or other systems does your work center around? How could you explain your research to a kindergarten student, a middle school student, a college student, a professional from a different field, or an expert from your field? You may note that some of these groups will require additional explanation to understand certain topics and ideas. What analogies could help explain those ideas that are not intuitive? Consider asking friends and colleagues outside of science to help you identify these points of confusion and ensure that your explanations and analogies are clear and interpretable. Note their questions, uncertainties, or misunderstandings, and adjust accordingly.

To concisely explain individual projects or papers you have published, consider outlining the study in bullet-points (including the background, methods, results, and implications). Once you've done this, determine which points are most integral to the story and must be included. What is the key message or takeaway from the study? What findings are most relevant to the airport bar-goer sitting beside you? For brevity and simplicity, it may be wise to leave out non-essential details such as findings not related to the study's central conclusion.

Once you've crafted this explanation, practice with friends or family members. Create space for them to ask questions and answer them in a respectful and thoughtful manner. Ask for feedback and adjust as needed to maximize comprehension. Once prepared, this summary can be used in a variety of settings beyond the airport bar, including your lab website or academic profile. For example, one author of this paper (S.B.), an Associate Professor at Georgia Tech, runs an engaging laboratory website that details the lab's research projects in clear and accessible language, incorporating striking videos and other visual aids.²⁴ Another great example comes from MIT Professor Ev Fedorenko, whose lab website provides clear summaries of individual research projects, publications, and research themes.²⁵ Keeping your laboratory website up-to-date with downloadable papers and project pages that summarize findings can enhance the accessibility of your work, help recruit students and fellows, inspire students into careers in science, or drive funding to your laboratory.

#2 - GET PUBLIC COMMUNICATION & ENGAGEMENT TRAINING

Effort/Time Investment: 2 - *One time commitment of <1 week.*

Potential Reach: *Groups 1-5. Science communication skills are useful in all settings.*

The process of developing effective communication skills is an ongoing, iterative journey that can be accelerated by effective training. Communications training can support your career by increasing engagement with your work. Whether you're speaking to colleagues at a conference, on a local radio program, or at a dinner table with friends or philanthropists, communication skills extend the value and impact of your research by facilitating understanding. We strongly encourage that aspiring science communicators obtain some level of communications training before initiating public messaging campaigns.

Before pursuing training, it is helpful to examine your personal goals and identify what outcomes you would ideally achieve through science communication. Since many trainings cater toward specific media formats (e.g., writing, video, audio) and to certain audiences (e.g., social media versus Scientific American) it is helpful to first consider who your ideal audience is, how you hope to influence them, and how you plan to reach them. To better understand this, consider: are there any formats where you will not feel comfortable (i.e., social media videos)? Is there a preferred medium through which you prefer to communicate (e.g., written versus spoken)? Is there a target audience you hope to reach, and if so, are there specific channels available to do so?

Based on your answers to these questions, there are many different trainings one may pursue. Graduate students may join the 3 Minute Thesis (3MT) competition, which challenges students to present their research in just three minutes to a lay audience, using only one slide. Early career researchers may also participate in science communication challenges, like the *BrainStorm Neuroscience Pitch Competition* held by the Mind Science Foundation, which offers \$100,000 in research funding to applicants who can effectively describe their proposals to non-specialist audiences. These innovative challenges incentivize students to sharpen their science communication skills by offering cash rewards to winners and provide infrastructure for engaging public audiences.

Additionally, researchers may enroll in specific science communication training. Many free or low-cost courses exist including those provided by Science Communication Lab,²⁶ EdX,²⁷ the University of Michigan,²⁸ the University of Colorado, Boulder,²⁹ and the Australian Broadcasting Corporation.³⁰ Additionally, students may wish to enroll in higher education programs offering formal training in science communication or science journalism, such as the specialized Master's degree programs offered at the University of California, Santa Cruz,³¹ Stony Brook University,³² Texas A&M University,³³ Massachusetts Institute of Technology,³⁴ or the University of North Carolina.³⁵

#3 ENGAGE WITH MEDIA & JOURNALISTS

Effort/Time Investment: 3 - Continuous commitment with occasional requirement for <1 day of work.

Potential Reach: *Groups 4 and 5: science enthusiasts and the general public.*

If not for the diligent work of science journalists, how would the public stay up to date on modern developments in science? Indeed, more than half of Americans rely on major news outlets to keep up with science.³⁶ Most pieces covering new research findings are written by journalists, who often conduct interviews with the scientists behind the work or other domain experts who provide additional context. Getting scientists to participate in interviews can be a challenge for journalists and writers, though it is imperative for accurate and comprehensive reporting. When asked for a quote or interview, researchers should consider engaging if comfortable, interested, and confident in their expertise as it relates to the subject. We acknowledge that speaking about controversial science topics may make media interactions more difficult; for this reason, you may wish to contact your university's public information officer (PIO) for guidance and assistance. It is also important to understand the journalist's and outlet's background (credibility, interest areas, etc.), which your PIO may also be able to help with.

When providing comments, it is advisable to prepare ahead of time and rehearse key points to facilitate a smooth interview and ensure that information is conveyed clearly. Your PIO may be able to help you prepare for these interactions. Becoming a go-to source for that reporter can save them time and trouble in preparing future articles and may generate opportunities for you to lend your expertise to traditional media. Building relationships with individual reporters is a fruitful step toward reliably contributing toward media articles. Additionally, you may wish to proactively speak with your institution's public relations department when you have new research findings to communicate, as they may be able to obtain media opportunities and help you craft your message for a non-specialist audience.

#4 CHALLENGE YOURSELF WITH PUBLIC SPEAKING

Effort/Time Investment: 3 - *Continuous commitment with occasional requirement for <1 day of work.*

Potential Reach: *Groups 4 and 5: science enthusiasts and the general public.*

Researchers at any level should challenge themselves by presenting to lay audiences, such as at schools, public-facing events or conferences, student organizations or clubs, or corporate groups. Additionally, events like science festivals,³⁷ Pint of Science,³⁸ and even "pop-up" science talks in public spaces offer opportunities to meet the public where they are. These invitations and events should be welcomed and accepted whenever possible and appropriate. Public-facing presentations are distinct from presentations at scientific or academic conferences which are delivered to peers. Rather, these public-facing opportunities offer chances not only to sharpen your science communication skills, but to facilitate a direct conversation between the public and your research topic. This is an opportunity to tell the broader

story of your work, explain its societal implications, understand any concerns your audience might have, and answer their questions. For instance, one author (B.R.) recalls speaking at a public event about his research on the neurobiology of autism, when he fielded a question about the role of vaccines. This created space for an open, respectful, and public discussion about the relationship between autism and vaccines, the existing data, and the history of this contentious topic. The conversation was polite, understanding, and seemingly impactful on both the questioner and the audience, highlighting the rare opportunity that public speaking allows for this type of exchange.

#5 AMPLIFY YOUR IMPACT ONLINE

Effort/Time Investment: 3 - 5 - *Continuous commitment requiring <1 day of work, <1 week of work, or consistent effort (depending on the type of activity)*

Potential Reach: *Groups 1 – 5. Virtual platforms can reach broad audiences.*

Billions of people worldwide (>60% of the global population)³⁹ use some form of social media every day, including over 90% of Americans.⁴⁰ This represents an unprecedented opportunity for science communication. Specifically, the “recommendation algorithms” used by platforms like TikTok and Instagram allow science content to reach users who may not typically seek it out. Thus, modern social media offers a robust new mechanism for engaging both science-interested audiences and the broader, uninitiated public. These platforms also offer venues for direct exchange with viewers through comments and other forms of inter-user messaging. The growing influence and dominance of social media apps may represent one of the greatest opportunities of all time for science communicators.

While these platforms may offer great upside, they are also unfortunately hosts to a great deal of misleading content. Misinformation and disinformation are highly prevalent on social media,⁴¹ and false claims also spread faster and farther than accurate information.⁴² This incentivizes accounts to post mis- and disinformation, which can be harmful for both science and society. As sources of high-quality information and expertise, researchers can contribute positively to our online information ecosystems by sharing openly about their research area, and working to address any misinformation spreading about their field. There are several formats and channels to engage the public. Here we will discuss these areas and their potential impact. However, we must first acknowledge that speaking about science online can spawn potential backlash and harassment. In today’s toxic online climate, scientists are often criticized and trolled.⁴³ This should be taken into consideration prior to engaging in any such efforts listed below. It is thus strongly encouraged to approach online discussions respectfully, especially around controversial topics. When met with opposing views, a friendly and polite demeanor will advance conversations much more effectively than hostility and condescension.

Social Media/Blogging: Sharing lay-accessible information online is a straightforward way to promote your work and draw in crowds that may not typically read scientific publications. Indeed, many scientists have used platforms like TikTok and Instagram to reach millions of users with science education.⁴⁴ However, not all social media efforts must target broad audiences by aiming for “virality;” there is also tremendous value in engaging smaller, more targeted groups, which can offer more intimate dialogue, as scientists can be viewed as a trusted messenger in their personal networks.⁴⁵ As researchers post about their work, they may naturally attract the attention of non-scientists with a penchant for learning. These users could include anyone from friends and relatives to total strangers. It can also be valuable for scientists to collaborate with content creators outside the science community. Building partnerships across communities allows scientists to serve as trusted resources when creators or their audiences have science-related questions. At the same time, creators outside the science space are already trusted messengers within their own communities; collaborating with scientists can therefore expand both the credibility and reach of scientific information.⁴⁶ For instance, one of the authors (M.F.) works with The Evidence Collective, a science communication nonprofit that has hosted events connecting science communicators with content creators in faith-based and parenting communities. As more scientists engage on social media, they should be intentional about reaching diverse audiences and building relationships beyond their immediate networks. When it comes to science communication, social media platforms like Instagram, TikTok, YouTube, X, Threads, Mastodon, BlueSky, Substack, LinkedIn and others are all viable options. However, these apps have different formats and cultures, which require different considerations.

- *Text-based platforms:* Many scientists use text-based platforms like X, Threads, LinkedIn, Mastodon, Substack, or BlueSky to share news from the lab and keep up on their colleagues’ work. Posts that summarize research papers can help popularize and disseminate publications well beyond the readership of primary journals and can also offer opportunities to generate excitement about new research findings and answer questions about the project. Furthermore, scientific papers promoted on Twitter receive significantly more downloads and higher Altmetric scores.⁴⁷ One author (H.E.) who uses LinkedIn for science communication has found that people from a variety of sectors are deeply receptive to science content when it is framed as an enabler of economic and societal value rather than as a narrow health topic. He has come to appreciate that science communication is less about broadcasting information and more about cultivating trust through consistency, humility, and responsiveness. In addition, he has found that his audience demands rigor and high-quality evidence, rather than overly simplified explanations. Scientists may also create blogs on platforms like *Substack*, which allow regular publication of newsletter-

like blogs to a self-selected audience of subscribers. One might start by simply talking about one's research or sharing science news with friends and family. *Video-based platforms:* Video-based platforms like Instagram, YouTube, and TikTok have become popular choices for science communicators. Creating video content may be more effortful and requires a unique skill set in video preparation and editing but has the potential to reach a broad range of users. These apps thus offer tremendous value for science communication efforts that meaningfully engage the public. Case studies illustrate the scalability of these digital platforms; for instance, one author (B.R.) has published several hundred educational videos on TikTok, averaging 54,000 views per post and exceeding 50 million total impressions.⁴ He encourages researchers to create videos about the scientific topics that excite them most, as enthusiasm is contagious. He also recommends framing the research through a lens that is relevant to a general audience. As one specific example, he posted a TikTok video in 2024⁴⁸ announcing a recent publication about empathy in online environments.⁴⁹ To make this broadly relevant, he framed the research around a culturally significant moment when actress Julia Roberts had been harassed by social media users about her appearance. By creating videos like these and engaging with viewers in the comments, researchers may connect with new audiences, propagate their research, build trust in scientists⁴ and potentially inoculate viewers against misinformation⁵⁰ and manipulation techniques.⁵¹ Considering the many available video-based social media platforms, it may be helpful to explore integrators such as HootSuite which allow for posting the same video across multiple social media platforms simultaneously.

Writing for Media

Scientists can partake in journalism by writing for media outlets directly. Many media outlets seek expert-written pieces, valuing the depth and credibility that researchers bring to their narratives. Moreover, articles written by researchers receive as much engagement as those written by reporters.⁵²

Contributing to media outlets offers a chance to hone new skills and craft impactful pieces that can attract talent, educate junior lab members, and promote your research. While assuming the role of a "writer" might seem daunting, these platforms often pair contributors with editors who offer guidance to refine and amplify the message. One notable platform welcoming expert contribution is The Conversation, known for its diverse coverage spanning policy, health, and history, all grounded in current academic research.⁵³

Outlets like The New York Times or The Atlantic are also open to pitches from scientific experts. For instance, one author of this paper (D.M.E.) penned an article for The Atlantic exploring the implications of brain science for our criminal justice system,⁵⁴ allowing for the weaving of neuroscience

data with broader societal issues. Publishing articles in such outlets not only enriches public discourse but also strengthens the impact of scientific research beyond academic circles.

#6 JOIN PODCASTS

Effort/Time Investment: 3 - *Continuous commitment with occasional requirement for <1 day's work.*

Potential Reach: *Groups 4 and 5, depending on the podcast scope and audience.*

Podcasts have exploded in popularity over the last decade, with the number of podcast listeners reaching 400 million people in 2022, up by over 100 million since just 2019.⁵⁵ Science podcasts have similarly grown exponentially in recent years, reaching broad public audiences.⁵⁶

Many podcasts are hosted by scientists or physicians, creating opportunities for deep dives on currently trending topics with added commentary. One author (R.R.), the host of the *Good Medicine*⁵⁷ podcast, notes that his audience appreciates deep conversations that really explore a subject even if they are not experts in that area. He believes his audience appreciates the earnest back and forth and the unpolished nature of the conversation, versus the performative, production-heavy nature of social media. He notes that preparation is key; the more time invested in learning about the interviewee, the more natural, effortless, and interesting the episode will be.

Podcasts from all sectors often invite experts to fill the other seat and discuss their research. These invitations represent valuable opportunities to exercise science communication. Sharing research in a conversational setting - rather than a traditional lecture approach - is an excellent opportunity to open your research area to audiences who typically may not interact with your materials. For instance, one author (A.M.) has appeared on podcasts focused on business, professional development and science, and found that preparation for the conversation is just as important as the conversation itself. Preparation can follow a three-step approach: first, understanding the host's perspective and background knowledge; second, considering the audience's likely level of familiarity with the topic; and third, integrating both into planned discussion points and communication strategy. Based on these factors, podcast guests can tailor their language, examples, and analogies accordingly. For example, in a podcast geared toward a diverse lay audience from multiple sectors, A.M. began with accessible analogies to frame the topic and then moved toward the practical impact of the work. Across interviews, she identified a small number of core messages she wanted listeners to remember and practiced delivering them concisely, knowing time might be limited.

Scientists should accept invitations to join podcasts as a guest whenever possible, and comfortable. Conversely, when scientists reject these invitations, podcasts may opt for alternative guest to discuss the same topic area. This is problematic because it encourages non-experts to speak on topics

beyond their expertise, leaving audiences without credible scientific voices. As a result, people may become more vulnerable to misinformation, which spreads more easily when individuals are unsure whom to trust or cannot access credible information.⁵⁸ By accepting podcast invitations, scientists may safeguard against the misrepresentation of our research fields.

One consideration when engaging with podcasts is to understand the unique perspective and potential agenda of the podcast. Every podcast is different, and many popular shows appeal to audiences who may be distrusting of science. It is thus important to understand the scope of the show and its target audience and be prepared with clear explanations, especially regarding controversial issues. Consider taking special precautions like planning key talking points that clarify confusing or divisive topics and cannot be misrepresented or taken out of context.

Conclusion:

In our ever-changing modern world, it's critical that scientists embrace a multifaceted and up-to-date approach to communicating our work. By leveraging science communication channels ranging from social media to podcasts and traditional media, scientists have many opportunities to engage and discuss their research with broad audiences. Science communication efforts help bridge the gap not only between scientists and the public, but also between academia and influential bodies like businesses and governments. These entities hold significant power in shaping policy, technology, and societal decisions.

When choosing to engage in science communication efforts, it's important that scientists opt for approaches that feel both *sustainable* and *authentic* to themselves. Scientists should feel comfortable and supported in setting boundaries to protect their mental health and personal safety. Communicators should also remain mindful of how their own backgrounds influence their messaging and, conversely, how diverse audiences perceive them. Finally, rather than viewing outreach as a simple lecture intended to teach or influence opinions, it should be seen as a shared process of meaning making between the creator and the viewer. Education is most successful when built upon a foundation of genuine exchange and respect.

Effective science communication is essential for building trust, addressing societal challenges, and ensuring that research benefits everyone. We, the churning gears of the scientific enterprise, must work diligently not only to ensure that the wheels of science continue to spin forward, but that the information gleaned from our efforts also reaches all people who benefit from it.

Declaration of Interests:

R.R. is a founder of Roon, a patient education app.

H.A.E. is an advisor to the Meadows Mental Health Policy Institute.

M.F. is on the Steering Committee for The Evidence Collective

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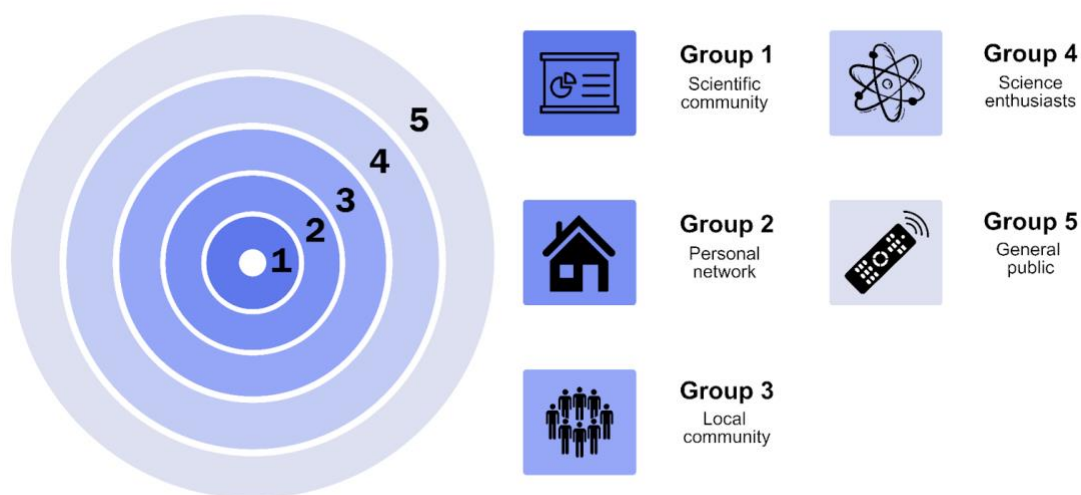


Figure 1: Potential Reach (PR) is a measurement (1-5) designed to gauge the audience each activity may reach and impact (**Fig. 1**). Higher scores indicate broader potential audiences. Groups are classified not by size, but by their level of accessibility to scientists. *Group 1* is composed of the *scientific community*: students, postdocs, professors and industry researchers who can be reached through traditional scientific outlets such as journal publications and conference presentations. Science communication efforts should ideally reach audiences this group. *Group 2* represents your *personal network*: first-degree connections who are not scientists, such as family and friends. Engaging Group 2 is the most accessible method of conducting science communication. *Group 3* represents your *local community*: second- and third-degree connections who you may not interact with on a regular basis. This may include neighbors, friends of friends, and other strangers you may speak with face to face. *Group 4* includes *science enthusiasts* who are entirely outside of your network. The only way to reach and address these individuals would be indirectly, through media channels such as podcasts or news outlets. Importantly, Group 4 comprises lay audiences who are interested in science and volitionally engage with science-related content. For example, joining a science-related podcast as a guest would reach this group, as the audience is self-selected for their interest in science. In contrast, *Group 5* comprises the *general public*: out-of-network individuals who are not explicitly interested in science. Thus, the only way to reach Group 5 is through media outlets that are not typically science focused, or by using tools such as social media that offer the ability to reach indiscriminate audiences.

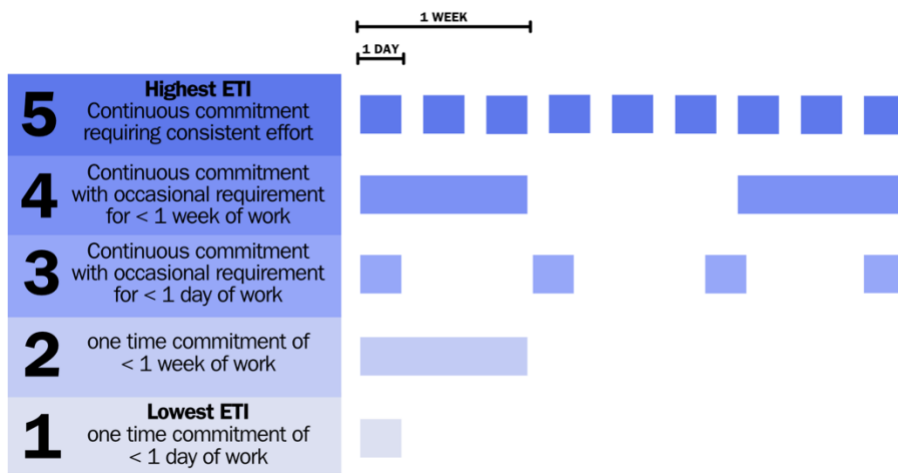


Figure 2: Effort/time investment (ETI) is a measurement (1-5) that gauges the relative level of effort and time each activity will require (**Fig. 2**). A value of 1 is the lowest, suggesting a one-time commitment of less than one day's work. A value of 5 is the highest, suggesting a continuous commitment that requires consistent effort. Several practices outlined here will only require a single, one-time commitment of ~1 day to ~1 week. These are represented by values of 1 and 2, respectively. For example, taking a 4-hour science communication workshop would be scored as an ETI of 1, as it is a one-time practice that requires less than one day of work. Traveling to a science communication conference for several days would be a 2: it's a one-time effort that could take up to a week's work. Other practices require a long-term commitment, that can demand an occasional investment of anywhere between one day's work and one week's work. These are represented by values of 3 and 4, respectively. Speaking occasionally with media outlets/organizations would be scored as a 3, as it is less than a day's work but is a task that's repeated periodically over a long-term commitment. Becoming a contributor to your local news outlet would be a 4; each article may take several days to write, and you are making a long-term commitment to contributing articles. Finally, practices may be rated as a 5 if they require continuous effort over a long-term commitment. An example of this is maintaining a social media channel or blog: these outlets must be regularly tended to and sustained.

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